

S&DS 365 / 665

Intermediate Machine Learning

Course Projects

Yale

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Course projects

- The course requires a project in place of a final exam
- Goal: hands-on ML experience on a problem you're interested in
- Doesn't need to be a big deal — do what interests you!

In line with the 4 P's of creative learning: Projects, peers, passion, and play (Mitch Resnick)

Project proposal

You must submit a *Project Proposal* to Gradescope.

- Deadline: Thursday, November 12, midnight
- Can work in groups of up to three people
- All people must be listed on the proposal
- Encouraged: Group size 2 or 3

Project proposal: Components

1. Project title
   What is a title for your project?
2. Group members
   Who is on your team? (Up to 3 people)
3. Problem statement
   What problem do you propose to study?
4. Data used
   What data sets (if any) will you use?
5. Machine learning methods
   What ML methods will your project explore?
6. Outcomes
   What are project goals? How will outcomes be evaluated?

Project writeup and submission

Each group will submit a short writeup:

- No longer than 3 pages; no use of AI for writing the text
- Required sections of main text will parallel the project proposals
- A supplement/appendix can be included — no restrictions on length or use of AI
- Hard deadline: Monday, December 21, 5:30pm

Project writeup: Main text ≤ 3 pages

1. Project title
2. Group members
3. Problem statement
4. Data used
5. Methods and approach
6. Results and conclusions
7. Statement on AI use

No use of AI allowed to write the main text

Project writeup: Supplement

- A supplement / appendix can (and should) be included
- Details of the project — Can include figures, charts, technical derivations, etc.
- No restrictions on use of AI to prepare the appendix

Assessment

Assessment of projects will be similar to assignments

If the project is submitted and shows significant effort full points will be given (15% of final grade)